FITNESS LANDSCAPE

WAFT | Evolutionary Terrain Analysis

What is a Fitness Landscape?

A **fitness landscape** is a visualization of how fitness varies across the space of possible genomes. Think of it as a terrain where:

- Height = Fitness
- Position = Genome configuration
- Peaks = High fitness (goals)
- Valleys = Low fitness (avoid)

Landscape Features

Peaks

High fitness regions. Evolution climbs toward these.

Valleys

Low fitness regions. Agents here die or mutate away.

Plateaus

Flat regions where mutation doesn't change fitness.

Ridges

Narrow paths connecting peaks. Hard to navigate.

Local vs Global Optima

Local Optimum: A peak that's higher than neighbors but not the highest overall. Evolution can get stuck here.

Global Optimum: The highest peak. The ultimate goal of evolution.

Escaping Local Optima

When stuck on a local optimum:

1. **Increase mutation rate** — Jump to new regions
2. **Use diversity maintenance** — Keep exploring different areas
3. **Add noise** — Random perturbations
4. **Change fitness function** — Reshape the landscape

Analyzing Your Landscape

```python
from waft.analysis import FitnessLandscape

landscape = FitnessLandscape(flight_recorder)

# Plot fitness over generations
landscape.plot_trajectory()

# Identify plateaus
plateaus = landscape.find_plateaus(min_length=5)

# Estimate landscape roughness
roughness = landscape.calculate_roughness()
```

Landscape Metrics

Metric	Meaning
Roughness	How much fitness varies with small changes
Correlation length	Distance over which fitness is predictable
Peak density	How many local optima exist
Basin size	How large attraction regions are

Visualization

```bash
# Generate landscape plot
waft analyze landscape --output landscape.png

# 3D surface (if 2D genome)
waft analyze landscape --3d

# Trajectory overlay
waft analyze landscape --trajectory
```

Multi-Objective Landscapes

When optimizing multiple fitness dimensions:

```python
landscape = MultiObjectiveLandscape(
    dimensions=["stability", "efficiency", "safety"],
)

# Find Pareto frontier
frontier = landscape.pareto_frontier()
```

Implications for Evolution

Landscape	Strategy
Smooth	Low mutation, hill climbing
Rugged	High mutation, exploration

Plateaued	Diversity, patience
Multi-modal	Population diversity

FITNESS LANDSCAPE | Know Your Terrain